Smart Water Logging ,Flood Prevention and Post Flood Event Management System

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Project Proposal Report

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DECLARATION

I declare that this is my own work, and this proposal does not knowingly incorporate any material previously submitted for a degree or diploma at any other university or higher education institution, nor does it contain any material that has been previously published or written by another person apart from where proper acknowledgment is given in the text.

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ABSTRACT

Urban flooding and water logging have become significant challenges in many cities due to rapid urbanization, inadequate drainage infrastructure, and increasing rainfall intensity caused by climate variability. Poorly monitored drainage systems often lead to blocked drains, rising water levels, and delayed responses from authorities, resulting in property damage, traffic disruption, and health risks for communities. Traditional drainage monitoring methods rely heavily on manual inspections and reactive measures, which are inefficient and unable to provide timely warnings.

Although several flood monitoring systems exist, many lack real time integration of multiple environmental parameters and predictive capabilities that can forecast potential flooding events before they occur. Additionally, limited visualization and alert mechanisms prevent authorities and stakeholders from responding effectively to early warning signs.

To address this gap, this research proposes a Smart Water Logging, Flood Prevention, and Post Flood Event Management System with a focus on drain water level prediction and real time drain monitoring. The system collects real time data from IoT sensors installed in drainage systems, including water level, flow rate, rainfall intensity, and blockage indicators. These data streams are transmitted to a cloud based platform where they are integrated with historical datasets to support predictive analysis. The system processes the data to detect abnormal conditions, assess flood and blockage risks, and forecast potential affected areas and timelines.

A web based dashboard provides interactive maps and visualizations to display real time drainage conditions and predicted risk levels. Automated alerts and notifications are generated when predefined thresholds are exceeded, enabling authorities to take preventive actions promptly. The system will be implemented using the MERN stack (MongoDB, Express.js, React, Node.js) for the web platform and Arduino based IoT sensors for real time data collection.

The expected outcome of this research is an intelligent monitoring and early warning system that improves flood preparedness, enhances urban drainage management, and reduces the social and economic impact of flooding. The proposed solution has potential social benefits by improving public safety and infrastructure management, while also offering commercial opportunities for smart city and environmental monitoring applications.

Keywords: **Smart Drain Monitoring, Flood Prediction, IoT Sensors, Real Time Monitoring, Flood Detection System**

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LIST OF ABBREVIATIONS

Abbreviation	Full Form
IoT	Internet of Things
API	Application Programming Interface
MERN	MongoDB, Express.js, React.js, Node.js
DB	Database
UI	User Interface
UX	User Experience
GPS	Global Positioning System
HTTP	Hypertext Transfer Protocol
HTTPS	Hypertext Transfer Protocol Secure
SMS	Short Message Service
ML	Machine Learning
SDG	Sustainable Development Goals

1. INTRODUCTION

1.1 Background and Context

However, in recent times, issues of urban floods and water logging have been on the increase in different parts of the world, especially in developing countries where rapid urbanization seems to be outgrowing infrastructure growth. Most drainage systems in these cities are either poorly managed or lack the ability to handle sudden increases in rainfall. When these drains are clogged or when there is unexpected rise in water levels, floods are likely to occur in a very short period of time.

For example, in Sri Lanka, flooding and water logging are common problems faced during periods of heavy rain, especially during the monsoon seasons. In cities such as Colombo and other densely populated areas, there is often an overflow of drainage due to blocked drains, intense rainfalls, and insufficient drainage capacity. As reported by the disaster management department and news sources, recent cases of heavy rainfalls in the country have led to flooding that has affected thousands of people and caused damage to infrastructure and transportation services.

The impact of water logging and drainage failure affects different groups of people, such as residents living in areas prone to floods, commuters who are affected by traffic congestion, and authorities in charge of drainage systems. When drainage systems are not monitored in real-time, authorities tend to act after the occurrence of floods.

There are several flood monitoring systems, which are already developed in various parts of the world. These systems are based on various parameters, such as weather forecasting, rainfall, satellite images, etc., to predict the flood situation. However, there are several disadvantages associated with the current systems used to monitor the flood situation. In some systems, only rainfall data are used, while the condition of the drainage system is not monitored. In some systems, the condition of the drains, such as the level of water, flow rate, and blockage of drains, are not monitored in real-time. In addition, some of the systems do not provide a visualization tool to take immediate actions before the flood situation occurs.

To overcome the shortcomings, this study proposes a Smart Water Logging, Flood Prevention, and Post-Flood Event Management System, specifically focusing on “Drain Water Level Prediction and Real-Time Drain Monitoring.” This system will employ Internet of Things sensors in drainage systems to collect real-time data, such as water level, flow rate, rainfall intensity, and possible blockages, and transmit this data to a cloud platform for storage and analysis along with historical data. Data analysis and prediction techniques will be applied to identify abnormal situations, estimate flood risks, and forecast areas and times that might be affected by floods. Web-based system tools will be used to display the data in a visually understandable format, along with automated alerts for authorities in case of abnormal situations.

This research is in line with the United Nations Sustainable Development Goals (SDGs). Specifically, this research contributes to the achievement of the following goals:

SDG 6 – Clean Water and Sanitation. This research aims to improve the management and implementation of water management and drainage systems.

SDG 11 – Sustainable Cities and Communities. This research contributes to the development and improvement of disaster resilience in the community.

SDG 13 – Climate Action. This research aims to improve the community's response to the threat of flooding due to climate change.

The project is also relevant to the research domains associated with Smart Cities, Internet of Things (IoT), Environmental Monitoring, and Data Analytics. Through the integration of IoT technologies and real time data processing and prediction analysis, the proposed system aims to contribute to the development of smart and more effective flood prevention strategies.

1.2 Literature Review

Flood prediction and urban drainage monitoring are two areas that have been extensively researched in the context of the growing number of extreme rainfall events and the rapid growth of urban areas. Various techniques such as hydrodynamic modeling, flood prediction using machine learning algorithms, Internet of Things (IoT) technology-based monitoring systems, and real-time early warning systems have been investigated in this context[1].

One of the most commonly used techniques in flood forecasting is hydrodynamic modeling, in which water flow is modeled using equations such as the Shallow Water Equations. Recently, a study presented a real-time high-resolution flood forecasting system using a GPU-accelerated hydrodynamic model, which is capable of running quickly for large-scale simulations in early warning systems [2]. Such systems can be used to obtain detailed flood maps and forecast the area affected by flooding in a very short period of time. However, this might be a problem in smaller systems because such models require high-performance systems and complex data sets.

Other studies have also been carried out on data-driven flood prediction methods, where machine learning and time series are used in the prediction of flood risk and water level based on historical rainfall and environmental data[2]. This method can be effective in improving the accuracy of the prediction results. However, the results can be poor in the event of unexpected environmental changes and the lack of historical data.

Another direction of research includes IoT-based flood monitoring systems. In this case, sensors such as ultrasonic water level sensors, flow sensors, and rainfall sensors are used in drainage systems to monitor environmental data in real time[3]. The data collected is then sent to a cloud server where it can be analyzed and visualized through a monitoring dashboard. IoT-based monitoring systems allow continuous observation of drainage systems, which helps in detecting abnormal conditions in a drainage system. However, most of the systems are based on monitoring, not prediction.

Besides academic studies, several flood monitoring systems have been developed for commercial and government purposes to support disaster management[4]. These systems can include weather forecasting data, hydrological modeling, and geographic information systems to support flood risk maps. Although these systems can be effective for large river catchments, they can be inadequate for monitoring the drainage system.

One limitation that has been identified with many of the existing systems is that they do not incorporate the integration of real-time IoT monitoring and prediction using flood analytics. In many cases, the systems have either used the simulation approach or the sensor monitoring approach without integrating the above two. In addition, some systems may require costly infrastructure and computational facilities that may not be feasible in developing countries.

In order to overcome the limitations identified in the previous research, the proposed research aims at developing a Smart Water Logging, Flood Prevention, and Post-Flood Event Management System based on the integration of IoT technology and data analysis. The system will be able to collect real time data on drainage systems such as the level of water, the rate at which the water is flowing, the intensity of the rainfall, and the blockage level. This real time data will be integrated with historical data in order to identify abnormal situations and predict flood events.

1.3 Research Gap

Flooding is becoming a major environmental and socio-economic problem in many nations, especially in tropical areas such as Sri Lanka. Flooding in urban areas and overflowing of drainage water are becoming a major problem in many tropical areas. Flooding in urban areas is mostly due to increased urbanization and climate change. In recent times, many areas in Sri Lanka have experienced severe flood disasters. For instance, Cyclonic Storm Ditwah (2025) is reported to have caused one of the most devastating flood disasters in Sri Lanka in recent times. Over 2.2 million persons in all 25 districts of Sri Lanka were affected, and over 100,000 houses were damaged, with hundreds of thousands of persons displaced from their homes.

Currently, flood monitoring and forecasting are mainly based on historical rainfall data, weather conditions, and hydrological models. Though these approaches are valuable for generating knowledge, they are often faced with several challenges. These challenges include the fact that most flood forecasting and monitoring systems are not able to monitor drainage conditions in real time, such that they cannot detect sudden blockages, increasing water levels, and abnormal conditions in drainage systems. Moreover, most forecasting models are based on only one parameter, such as rain intensity or river level, which makes it hard to make accurate predictions in the presence of abnormal local drainage system failures.

Another weakness of these approaches is the lack of integration between the results of IoT based monitoring systems and weather forecasts. Some approaches are based only on monitoring systems, while others are based only on weather forecasts. This is not enough to cover the complexity of urban flooding, since both drainage conditions and weather forecasts are involved in the process of urban flooding.

Thus, there is a major research gap in designing hybrid flood prediction systems that incorporate IoT sensor data and weather forecast data. The proposed research aims at filling this gap by designing a hybrid flood prediction model that incorporates IoT sensor data and weather forecast data for drain monitoring and flood prediction. For instance, it will incorporate real-time data related to water level, flow rate, rainfall intensity, and presence of blockages in the drains. It will also incorporate weather forecast data and historical flood data.

Through the use of multiple data sources, which are integrated by the proposed system, it is possible to achieve an earlier detection of abnormal drainage situations, improve flood prediction accuracy, and generate alerts in a timely fashion for authorities and communities. This method represents a more comprehensive and intelligent way to deal with flood situations, particularly in urban drainage systems that are susceptible to sudden flooding under extreme weather conditions.

2. OBJECTIVE

2.1 Main Objective

The main objective of this research is to design and develop a hybrid flood prediction and real time drain monitoring system that integrates IoT sensors.

2.2 Specific Objectives

- To design and implement an IoT based drainage monitoring system that collects real time data such as water level, flow rate, rainfall intensity, and blockage indicators using sensors connected to an Arduino microcontroller.
- To develop a data integration mechanism that combines real time IoT sensor data with historical drainage data and weather forecast information to support accurate flood risk prediction.
- To design and implement a hybrid flood prediction model that analyzes drainage conditions and weather forecast data to identify abnormal situations and estimate possible flood risks and timelines.
- To develop a web based dashboard using the MERN stack to visualize real time drainage data, predicted flood risks, and affected areas through interactive maps and graphical analytics.
- To implement an automated alert and notification system that sends warnings when predefined thresholds or predicted flood risks are detected in the drainage system.
- To evaluate the effectiveness and accuracy of the proposed system in detecting abnormal drainage conditions and predicting potential flooding events.

3. METHODOLOGY

3.1 Research Approach

This study follows the Design Science Research (DSR) method with an emphasis on the experimental evaluation. This method is appropriate for the study because the objective of the research is to design and develop a technological solution, which in this case will be the Hybrid Flood Prediction and Real-Time Drain Monitoring System. This technological solution will be one that combines the use of IoT technology in the monitoring of the drainage system with predictive analysis to identify any abnormalities and potential cases of flooding.

The use of the experimental method will be to evaluate the effectiveness and accuracy of the designed system, which will be done by simulating the data obtained from the drainage system and real-time sensor data, as well as the use of rainfall data.

3.2 Technical Methods and Algorithms

The data collection process for the proposed system will be done in real time, and the data will be collected using IoT sensors located in the drainage channels. These sensors will collect data on water levels, flow rates, rainfall, and blockages, if any. This data will then be sent to the system through a microcontroller.

The hybrid model of flood prediction will be achieved by integrating data obtained from the IoT sensors with weather forecast data and historical data on flood occurrences. This will enable the system to analyze the data, identify any abnormalities, such as rising water levels, decreasing flow rates, and high rainfall, among others.

The abnormalities will be detected through basic statistical analysis and threshold based algorithms. Moreover, the system will also be able to predict the changes in water levels over time, which might cause flooding, using the time series prediction method. This will be achieved by integrating data obtained from the IoT sensors with weather forecast data, which will enable the system to estimate the levels of flooding and the time of the flood.

3.3 Tools and Platforms with Justification

The system will be developed using the MERN stack along with IoT hardware components.

- **Arduino Microcontroller** – Used to collect data from sensors and transmit it to the server due to its low cost, reliability, and ease of integration with IoT sensors.
- **IoT Sensors** – Water level sensors, flow sensors, rainfall sensors, and blockage detection sensors will be used to monitor drainage conditions in real time.
- **MongoDB** – Used to store sensor data, historical records, and prediction results due to its ability to handle large volumes of real-time data efficiently.

- **Express.js and Node.js** – Used to develop the backend API for data processing, prediction algorithms, and communication between sensors and the database.
- **React.js** – Used to build a responsive web-based dashboard that visualizes drainage data, flood predictions, and alerts.
- **Leaflet.js or Mapbox** – Used to create interactive maps to display monitored drainage locations and flood risk areas.

3.4 Data Collection and Ethical Considerations

Data for this research will be collected from two main sources:

- **Real time sensor data** collected from IoT devices placed in the drainage systems.
- **Secondary data sources** include historical rainfall data, flood data, and weather forecast data, which can be collected from various sources such as public meteorological services or public data sources.

Ethical considerations of this research will be maintained in the following ways: No personal information will be collected in this research. Only environmental and infra related information will be collected and monitored. Moreover, the collected data will be utilized solely for research and development purposes.

3.5 Validation Strategy and Performance Metrics

The proposed system will be validated through a series of experimental tests and performance evaluations. The system's prediction accuracy will be tested by comparing predicted flood risks with known drainage behavior patterns or simulated flooding scenarios.

Several performance metrics will be used to evaluate the effectiveness of the system, including:

- **Prediction Accuracy** – Measures how accurately the system predicts potential flooding events.
- **Response Time** – Measures the time taken for the system to detect abnormal conditions and generate alerts.
- **Sensor Data Reliability** – Evaluates the consistency and stability of the collected sensor data.
- **Alert Effectiveness** – Measures how efficiently the system generates warnings when thresholds are exceeded.

4. HIGH-LEVEL SYSTEM ARCHITECTURE

4.1 Overview of the Proposed Solution

The proposed solution is a Hybrid Flood Prediction and Real-Time Drain Monitoring System that monitors drain conditions and predicts possible flood occurrences. The proposed solution combines IoT sensors, real time processing, weather forecast, and a web based interface to facilitate flood prediction and decision-making.

The architecture consists of four main layers:

- Data Collection Layer (IoT Sensors)
- Communication Layer
- Data Processing and Prediction Layer
- Visualization and Alert Layer

IoT sensors are placed in drain systems to collect environmental parameters such as water level, water flow, rain intensity, and blockage. Collected parameters are sent to a server where they are processed and combined with forecast and historical data to produce flood prediction results. The results are then displayed on a web based interface to produce alerts when abnormal conditions are present.

4.2 High-Level System Diagram

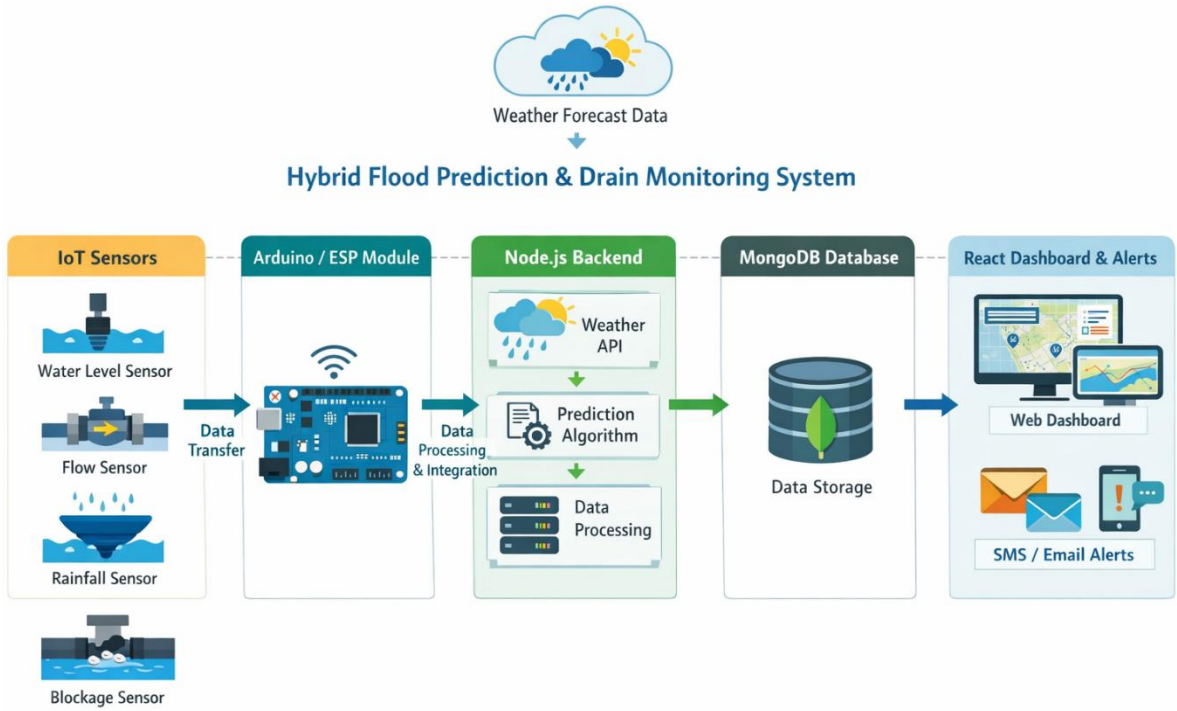


Figure 1High level system diagram

4.3 Individual Components and Their Parts

1. IoT Sensor Layer

This layer is responsible for real-time environmental data collection from drainage systems.

Components include:

- Water Level Sensor – Measures the water level inside the drain.
- Flow Rate Sensor – Measures the speed of water flow to detect abnormal flow conditions.
- Rainfall Sensor – Measures rainfall intensity in the monitored area.
- Blockage Detection Sensor – Identifies potential drain blockages based on flow and water level patterns.

These sensors are connected to an Arduino microcontroller, which collects and transmits the data.

2. Communication Layer

The communication layer is responsible for transmitting sensor data to the backend system.

Components include:

- Arduino Microcontroller
- Wi-Fi / Internet communication module

The microcontroller processes sensor readings and sends them to the server using HTTP or MQTT protocols.

3. Data Processing and Prediction Layer

This layer performs data storage, analysis, and prediction.

Key components:

- Node.js Server
- Prediction Algorithm
- Weather API Integration
- MongoDB Database

Functions:

- Store real-time and historical sensor data
- Integrate weather forecast data
- Detect abnormal drainage conditions
- Predict flood risk levels and timelines

The hybrid prediction model combines:

- Real-time sensor data
- Historical flood data
- Weather forecast information

4. Visualization and Alert Layer

This layer presents the processed information to users and generates warnings.

Components include:

- React Web Dashboard
- Interactive Map (Leaflet.js / Mapbox)
- Alert Notification System

Functions:

- Display real time drainage conditions
- Show predicted flood risk levels
- Visualize monitored locations on maps
- Generate alerts when thresholds are exceeded

Alerts may be sent through:

- Email notifications
- SMS alerts
- Dashboard warnings

4.4 Component Interactions

The system components interact in the following sequence:

- IoT sensors collect drainage data.
- Arduino reads the sensor values.
- Data is transmitted to the Node.js backend server.
- The server stores the data in MongoDB.
- The backend integrates weather forecast data.
- Prediction algorithms analyze the combined data.
- Results are sent to the React dashboard.
- If flood risk thresholds are exceeded, the system generates alerts.

4.5 Data Flow Explanation

The data flow within the system follows these steps:

1. Sensor Data Acquisition - Sensors collect environmental parameters from the drainage system.
2. Data Transmission - Arduino sends sensor readings to the backend server through an internet connection.
3. Data Storage - The Node.js server stores incoming data in the MongoDB database.
4. Data Processing and Prediction - The system analyzes real time sensor data together with weather forecast data and historical records.
5. Visualization - Processed results are displayed on the web dashboard with charts and maps.
6. Alert Generation - If abnormal drainage conditions or predicted flood risks are detected, the system automatically generates alerts

4.6 Scalability and Security Considerations

Scalability

The system is designed to support future expansion by:

- Adding multiple sensor nodes across different drainage locations
- Using cloud based database storage
- Implementing modular backend services

This allows the system to monitor large urban drainage networks.

Security

Several security measures will be implemented:

- Secure API communication (HTTPS) to protect data transmission
- Authentication and authorization for dashboard access
- Input validation and data encryption to prevent unauthorized access
- Secure database access controls

5. USER REQUIREMENTS

5.1 Stakeholder Identification

- Municipal Authorities / Local Government - Responsible for managing urban drainage systems and responding to flood risks. They will use the system to monitor drainage conditions and receive early warnings.
- Disaster Management Authorities - Organizations responsible for coordinating emergency responses during flood events. The system will support them with real time flood risk information and alerts.
- Urban Planning and Infrastructure Departments - These stakeholders can use the collected data to analyze drainage system performance and improve future urban infrastructure planning.
- Community Residents - Residents living in flood-prone areas benefit indirectly through early warnings and improved flood prevention measures.
- System Administrators - Responsible for maintaining the system, managing users, monitoring sensor connectivity, and ensuring the proper functioning of the platform.

5.2 Functional Requirements

- The system shall collect real time data from IoT sensors installed in drainage systems, including water level, flow rate, rainfall intensity, and blockage indicators.
- The system shall transmit sensor data from the Arduino microcontroller to the backend server through an internet connection.
- The system shall store real time and historical drainage data in a database.
- The system shall integrate weather forecast data from external weather APIs.
- The system shall analyze collected data to detect abnormal drainage conditions such as rising water levels or potential blockages.
- The system shall generate flood risk predictions using a hybrid prediction approach that combines IoT sensor data and weather forecast information.
- The system shall display real time monitoring data and prediction results on a web based dashboard.
- The system shall visualize monitored drainage locations and flood risk levels using interactive maps.
- The system shall automatically generate alerts when predefined thresholds or predicted flood risks are detected.
- The system shall allow administrators to monitor system status and sensor connectivity.

5.3 Non-Functional Requirements

1. Security

- The system shall ensure secure communication between sensors, server, and users using HTTPS protocols.
- Only authorized users shall be allowed to access the dashboard and system management features.
- Data stored in the database shall be protected from unauthorized access.

2. Performance

- The system shall process and update real-time sensor data with minimal delay.
- Alerts shall be generated immediately when abnormal conditions are detected.
- The system shall handle multiple sensor nodes simultaneously.

3. Usability

- The web dashboard shall provide a simple and user friendly interface for monitoring drainage conditions.
- Data visualization shall include graphs, charts, and interactive maps for easier understanding.

4. Reliability

- The system shall ensure consistent data collection from sensors.
- The system shall detect and report sensor failures or communication interruptions.

5. Scalability

- The system shall support the addition of new sensor nodes and monitoring locations without major system modifications.

6. Maintainability

- The system architecture shall be modular to allow easy updates, debugging, and maintenance.

6. COMMERCIALIZATION PLAN

6.1 Target Market and Customer Persona

The main target market for the proposed system will be the various municipal councils and disaster management departments that are in charge of managing drainage systems in the country. These departments need effective means of monitoring the drainage systems in order to act in a timely manner in the event of any possible flooding occurrences.

Other than the main target market, the system will also be beneficial to the various city residents and businesses that fall within the flood-risk areas in the country.

Persona 01

Table 1 persona 01

Name	Sanduni Fernando
Demographic	34 years old, Disaster Management Officer
Goal	Receive early warnings about possible floods and coordinate emergency response
Bio	Sanduni works at a regional disaster management office and is responsible for coordinating flood alerts and evacuation plans. She relies on accurate data to make quick decisions.
Pain Points	Flood warnings often arrive too late, limited real-time data from drainage systems
Motivations	Protect communities, improve disaster preparedness, reduce casualties during floods

Persona 02

Table 2 persona 02

Name	Nimal Perera
Demographic	42 years old, Municipal Engineer, Colombo Municipal Council
Goal	Monitor drainage systems in the city and detect flood risks early
Bio	Nimal is responsible for managing drainage infrastructure in urban areas. He needs reliable real time information about water levels and drain blockages to respond quickly during heavy rain.
Pain Points	Lack of real time drain monitoring, delayed flood information, difficulty locating blockages quickly
Motivations	Reduce flood damage, improve city infrastructure management, respond faster to emergencies

Persona 03

Table 3 persona 03

Name	Ruwan Silva
Demographic	38 years old, Small Business Owner living in a flood prone urban area
Goal	Receive early alerts about flooding to protect his property and family
Bio	Ruwan owns a small grocery shop in a low lying area that frequently floods during heavy rain. Early warnings would help him move goods and take precautions.
Pain Points	Floods happen suddenly, property damage, lack of early warnings
Motivations	Protect business assets, ensure family safety, prepare before flooding occurs

6.2 Value Proposition

The proposed system will help in real time monitoring, early flood prediction, and generation of alerts through IoT sensor technology and data analytics. It will help authorities take preventive measures before a flood occurs.

The proposed solution will help in improving urban flood management, safety, and disaster preparedness, and also help in reducing damages to infrastructure, agriculture, and properties.

6.3 Revenue Model

The system can generate revenue through several approaches:

1. **Government and Municipal Contracts** for installing and maintaining the monitoring system in urban areas.
2. **Subscription based monitoring services** where authorities pay for access to the monitoring dashboard and predictive analytics.
3. **IoT hardware installation packages** that include sensors, system setup, and maintenance.

6.4 Cost Estimation

The main costs involved in developing and deploying the system include:

- **Hardware costs** such as IoT sensors, microcontrollers, communication modules, and installation materials.
- **Software development costs** for building the data processing system and web-based monitoring dashboard.
- **Cloud infrastructure and data storage costs** for managing real time sensor data and analytics.
- **Maintenance costs** for sensor calibration, system updates, and technical support.

6.5 Pricing Strategy

The pricing strategy can follow a service based model where municipalities pay for system installation and ongoing monitoring services.

Possible pricing structure includes:

- **Initial system installation fee** for IoT sensor deployment.
- **Monthly or annual subscription fees** for using the monitoring dashboard and receiving predictive analytics and alerts.
- **Maintenance and support packages** for long term system operation.

6.6 Intellectual Property Considerations

The system design, data integration approach, and predictive algorithms can be protected through **software copyrights and potential patents** for the integrated monitoring and prediction framework. Proper licensing and intellectual property protection will help ensure that the developed technology can be commercially deployed while protecting the research contributions.

7. BUDGET AND JUSTIFICATION

7.1 Hardware Cost

The hardware components are required to build the IoT-based drainage monitoring prototype. These components enable the system to collect real-time environmental data from drainage systems.

Hardware Component	Quantity	Estimated Cost (LKR)	Purpose
Arduino Uno (existing)	3	6300	Microcontroller used to read sensor data and control communication
Water Level Sensor (Ultrasonic)	3	750	Measures the water level in the drainage system
Water Flow Sensor	3	2970	Measures the flow rate of water in the drain
Rainfall Sensor	3	900	Measures rainfall intensity in the monitored area
ESP8266 WiFi Module	3	4050	Enables internet connectivity to send data to the server
Jumper Wires and Breadboard	3 set	2400	Used to connect sensors and components
Power Supply / Adapter	3	3,000	Provides stable power for the system
Miscellaneous Components	-	1,200	Additional connectors, casing, and installation materials

Table 4Hardware cost

7.2 Software Licenses

The proposed system will mainly use **open source technologies**, which significantly reduces software costs.

Software Tool	License Cost	Purpose
NeonDB	Free (Community Edition)	Database for storing sensor and prediction data
Node.js & Express.js	Free	Backend server development
React.js	Free	Frontend dashboard development
Arduino IDE	Free	Programming and controlling the microcontroller
Leaflet.js / Mapbox	Free / Basic free tier	Interactive map visualization

Table 5Software Licenses

7.3 Cloud Services

Cloud services may be used to host the backend system and database for real-time data access.

Cloud Service	Estimated Cost	Purpose
Cloud Hosting (Vercel / Render / Railway)	Free Tier	Hosting the web application
NeonDB Cloud Database	Free Tier	Cloud-based database for storing system data
Weather API	Free Tier	Access to real-time weather forecast data

Table 6Cloud Service

7.4 Human Effort Cost

The development of the system requires effort from the research team in system design, implementation, and testing. Although this project is developed as part of an academic research project, the estimated human effort cost is included to represent the potential development cost in a commercial scenario.

Task	Estimated Effort	Estimated Cost (LKR)
System Design and Planning	30 hours	30,000
IoT Hardware Development	40 hours	40,000
Backend Development	50 hours	50,000
Frontend Dashboard Development	40 hours	40,000
Testing and Evaluation	30 hours	30,000

Table 7Human Effort Cost

7.5 Justification for Each Expense

- The hardware costs are required in order to build the IoT based prototype that can collect real time data on drainage. Sensors like water level sensors, flow sensors, and rain sensors can be used in order to accurately sense the environment.
- The software tools are required in order to build the web based monitoring platform and prediction tool. Open source technologies can be very cost effective in reducing costs while at the same time delivering robust development capabilities.
- The cloud services are required in order to host the developed application and store data from sensors in real time.
- The cost of human effort represents the effort required to develop the system. Although the cost is not incurred in the academic project, it represents the development effort required for the actual implementation of the system.
- It may be concluded that the proposed system is cost effective and suitable for the smart city flood monitoring system.

8. WORK BREAKDOWN STRUCTURE (WBS)

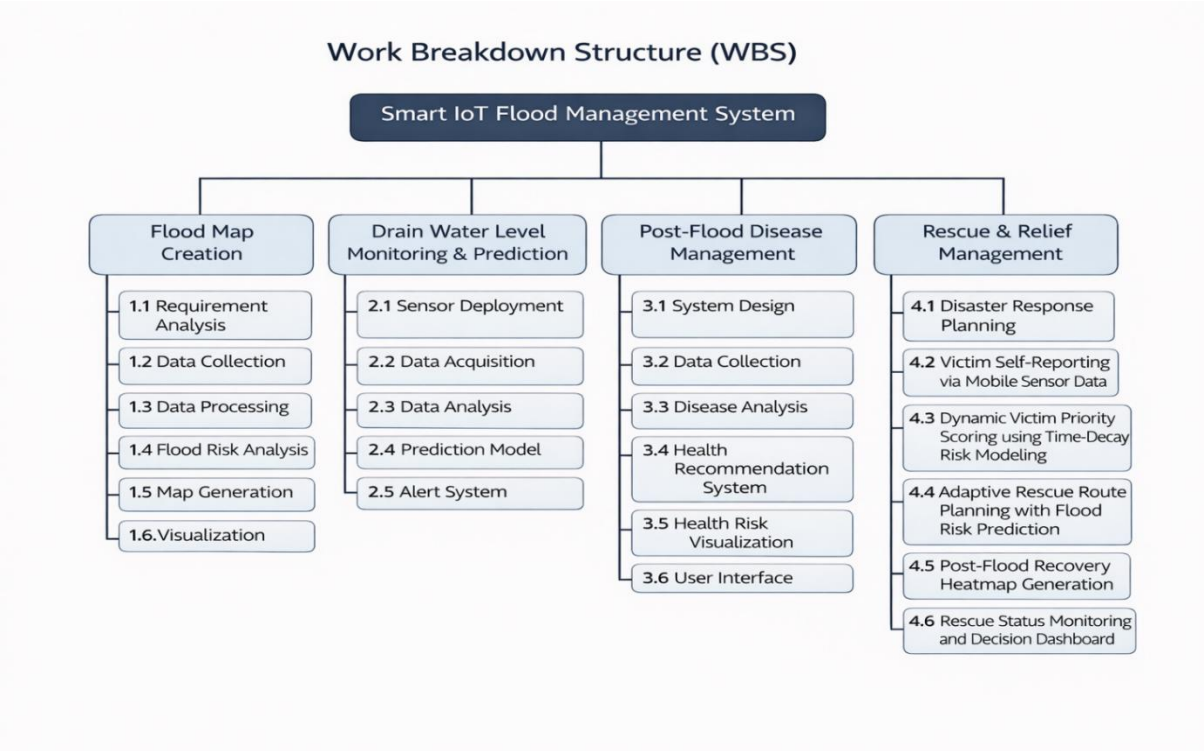


Figure 2 WBS

9. GANTT CHART

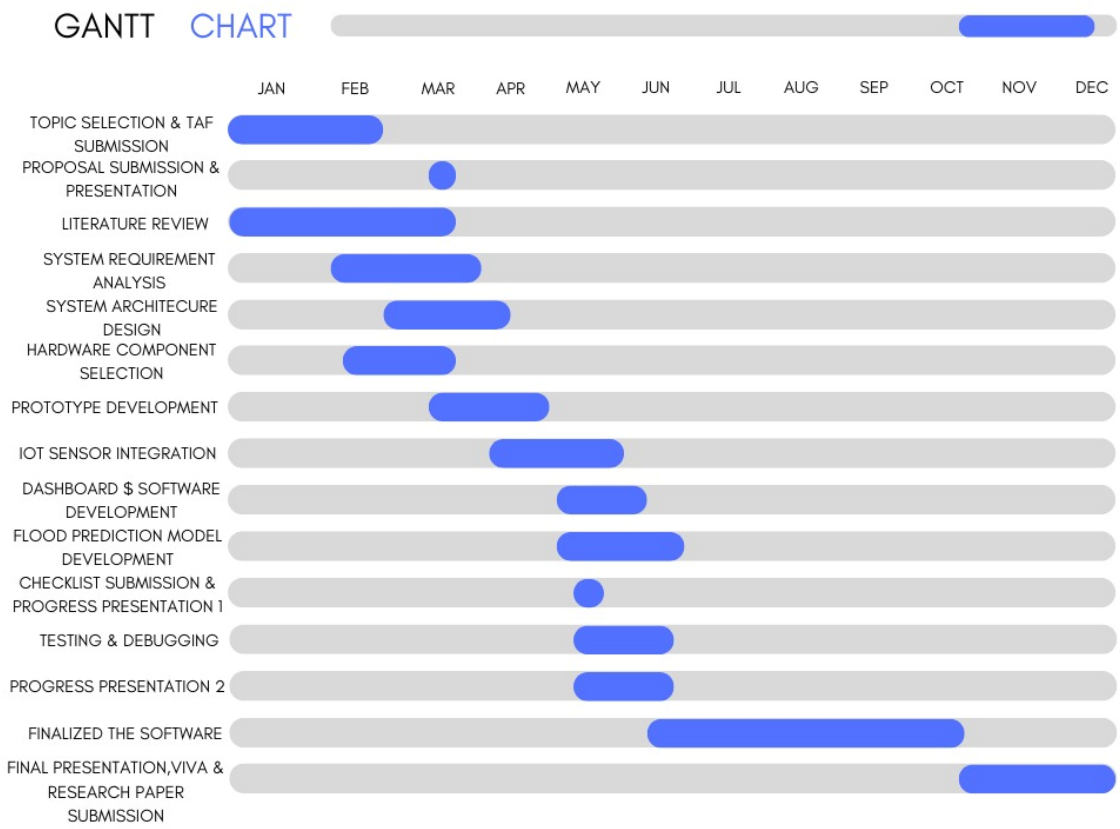


Figure 3 Gantt Chart

10.REFERENCES LIST

[1] S. Khosh Bin Ghomash, S. Deng, and H. Apel, “Enabling real-time high-resolution flood forecasting for the entire state of Berlin through multi-GPU accelerated physics-based modeling,” *Natural Hazards and Earth System Sciences*, vol. 26, no. 1, pp. 85–101, Jan. 2026, doi: 10.5194/nhess-26-85-2026.

[2] J. Henonin, B. Russo, O. Mark, and P. Gourbesville, “Real-time urban flood forecasting and modelling - A state of the art,” *Journal of Hydroinformatics*, vol. 15, no. 3, pp. 717–736, 2013, doi: 10.2166/hydro.2013.132.

[3] W. F. Krajewski *et al.*, “Real-time flood forecasting and information system for the state of Iowa,” *Bull. Am. Meteorol. Soc.*, vol. 98, no. 3, pp. 539–554, Mar. 2017, doi: 10.1175/BAMS-D-15-00243.1.

[4] L. Wang *et al.*, “A review of the flood management: from flood control to flood resilience,” Nov. 01, 2022, *Elsevier Ltd*. doi: 10.1016/j.heliyon.2022.e11763.